Analysis of Student Dropout

Causes and Prediction

Prepared by: *Yogendra Prasad Luitel (M. ED English)*

Course: *Data Analysis / Python/Power BI*

Date: *(2026/3/14)*

1. Introduction

Student dropout is a major challenge in educational systems worldwide. Students leaving studies early affect both their personal growth and institutional performance. Various factors such as financial constraints, academic difficulties, family background, stress, and lack of internet access influence dropout. In Nepal, student dropout is also a significant issue impacting education quality. This study aims to identify key factors affecting dropout and analyze the relationship between student characteristics and dropout behavior. Another objective is to develop a predictive model to forecast student dropout likelihood. The findings will help institutions recognize at-risk students and take preventive measures. Policymakers can use insights to design targeted programs. This research focuses on data-driven analysis using student features available in the dataset. Limitations include the exclusion of qualitative factors like personal motivation or school environment.

1.2 Research Objectives

The main objectives of this study are:

1. To identify the key factors influencing student dropout.
2. To analyze the relationship between student characteristics (age, gender, family income, study habits, stress, etc.) and dropout behavior.
3. To develop a predictive model that can forecast the likelihood of a student dropping out.

1.3 Significance of the Study

- Help educational institutions identify at-risk students and implement preventive measures.
- Assists policymakers in designing targeted programs to reduce dropout rates.
- Provides insight into the main factors influencing dropout, which can guide interventions at both academic and social levels.

1.4 Scope and Limitations

- The study focuses on analyzing student dropout patterns using the provided dataset of student characteristics.

- Prediction is limited to the features available in the dataset; external factors not included may also influence dropout.
- This analysis is primarily data-driven and may not capture qualitative factors such as personal motivation or school environment.

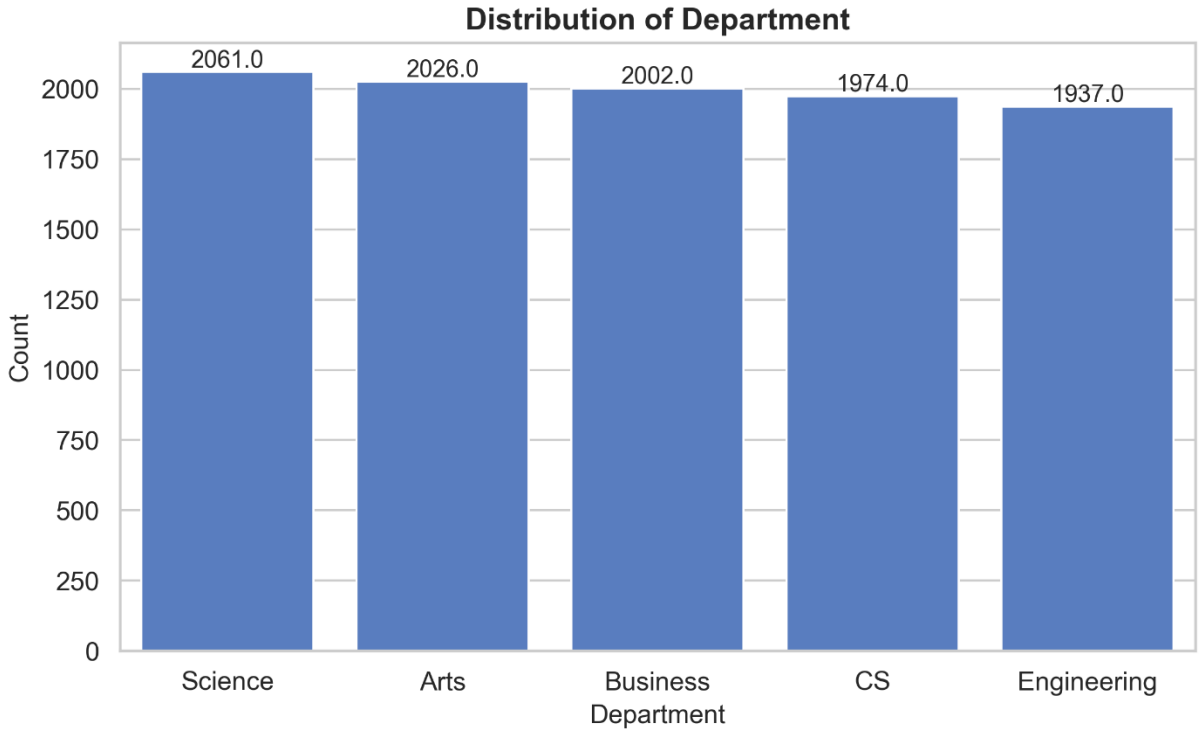
2. Literature Review

Student dropout has been widely studied in educational research. Many studies identify financial difficulties, low academic performance, and lack of family support as main causes. Other factors include part-time jobs, long travel time, and limited access to learning resources. Stress and mental health issues are also strongly linked to dropouts. Previous research highlights that student motivation and engagement reduce dropout risk. Machine learning techniques such as Logistic Regression, Decision Trees, and Random Forests have been used for dropout prediction. Studies show that predictive models can identify at-risk students effectively. Data-driven analysis helps institutions take preventive actions early. However, most studies focus on developed countries, and limited research exists in the Nepalese context. This research aims to fill that gap by analyzing student dropout patterns using local data.

3. Data Description

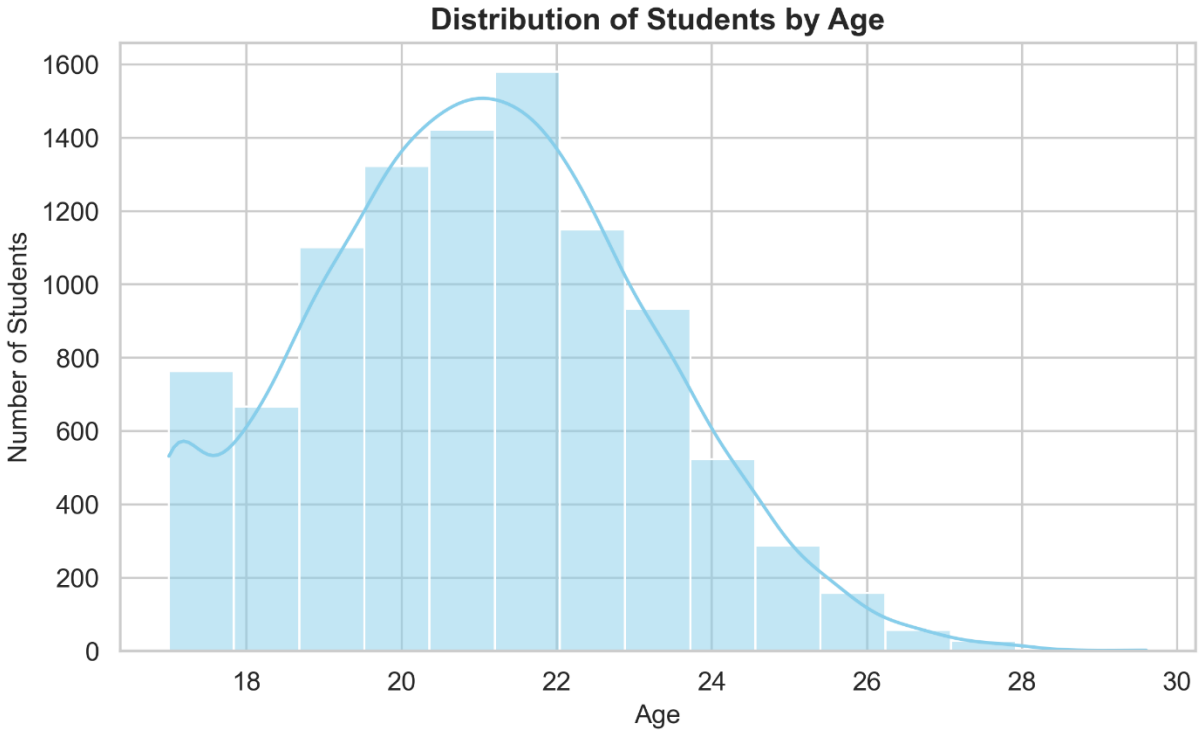
The data set used in this study contains records of students with various characteristics. It includes 18 features such as Student.Id Age, Gender, Family Income, Internet Access, Study Hours, Attendance Rate, and Stress Index. The target variable is **Dropout**, indicating whether a student left their studies. Some columns initially had missing values, which were handled through imputation. Categorical features like Gender, Internet Access, Part-Time Job, and Scholarship were encoded into numeric values for analysis. Numeric features such as Family Income, Study Hours, and Stress Index were analyzed using descriptive statistics. The dataset provides a mix of demographic, academic, and social information. This structure allows exploration of both personal and environmental factors affecting dropout. It is suitable for both exploratory analysis and predictive modeling. Overall, the dataset provides a comprehensive view of student behavior and risk factors.

Figure 2: Confusion matrix of Logistic Regression predictions



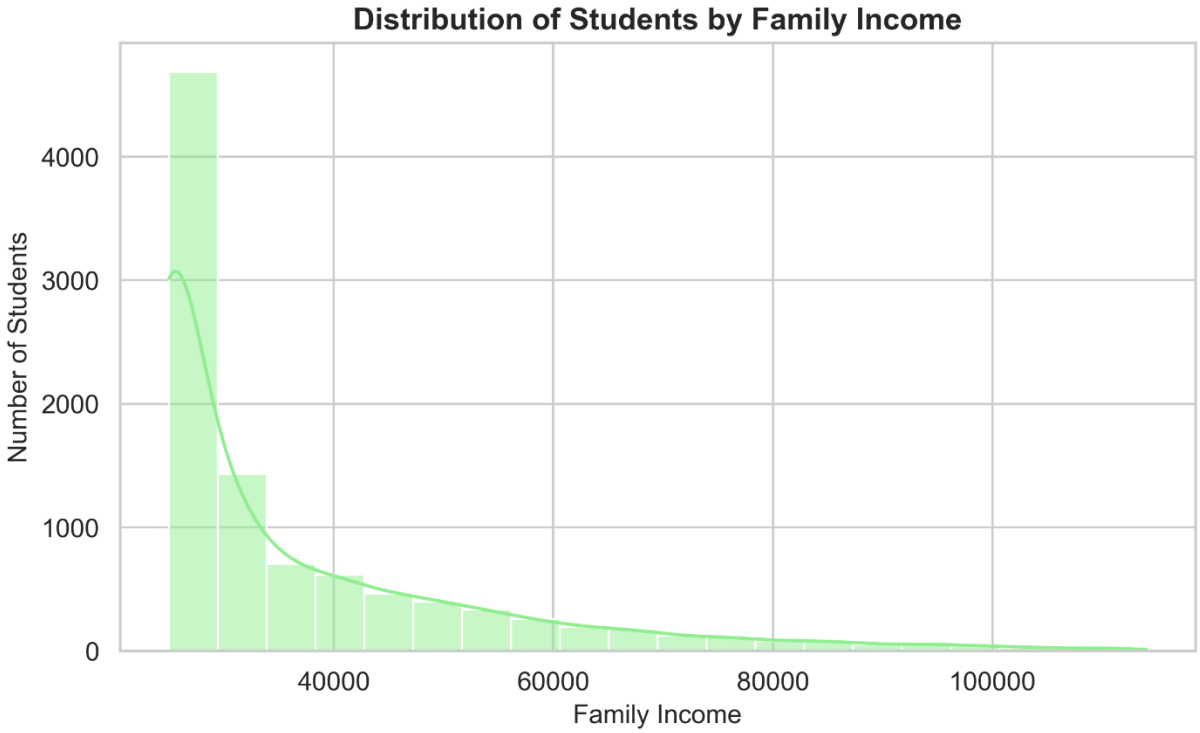
The bar chart shows the distribution of students across different departments. The highest number of students belongs to the **Science** department, followed by **Management** and **Education**. Departments like **Humanities** and **Others** have relatively fewer students. This uneven distribution indicates that certain fields attract more enrollment than others. Understanding department-wise distribution helps analyze dropout patterns and plan targeted interventions for underrepresented departments.

Fig 2: Distribution of Students by Age



The histogram illustrates the distribution of students' ages in the dataset. Most students concentrate between 18 and 22 years, which indicates typical college-age enrollment. A smaller number of students are below 18 or above 25, forming minor tails in the distribution. The KDE curve highlights a single peak around 20 years, suggesting that the majority of students are in their early college years. Analyzing this age distribution is important, as it helps in understanding dropout trends across different age groups and identifying at-risk students.

Fig 3: Distribution of Students by Family Income



The histogram depicts the distribution of students' family income in the dataset. Most students come from families with moderate income levels, while very low and very high incomes are less frequent. The KDE curve shows a right-skewed distribution, indicating that fewer students belong to high-income families. Income disparities may influence dropout rates, as students from lower-income households may face financial challenges. Understanding family income distribution helps identify at-risk groups and supports planning targeted interventions to reduce dropouts.

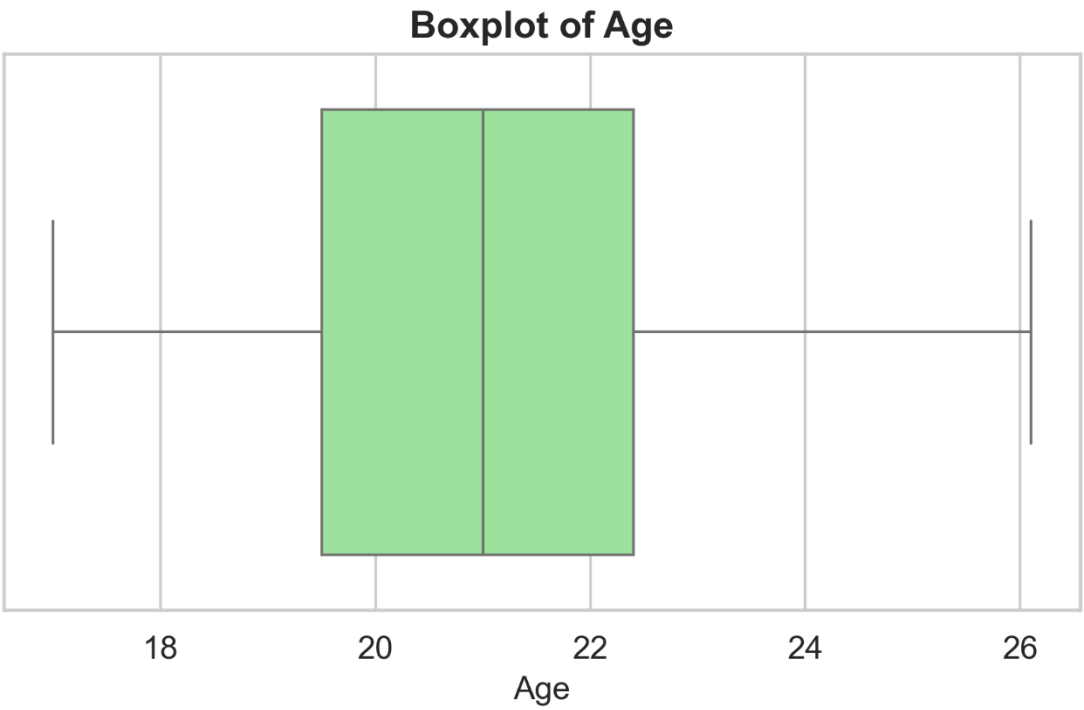
4. Data Preprocessing

Data preprocessing is an essential step before analysis and modeling. Missing values in columns like Family Income, Study Hours, Stress Index, and Parental Education were filled using median or mode. Categorical variables such as Gender, Internet Access, Part-Time Job, Scholarship, and Dropout were encoded into numeric values. Outliers in numeric features were checked and handled where necessary. Features were scaled if required to improve model performance. The dataset was split into training and testing sets for predictive modeling. Data consistency and integrity were verified after cleaning. Preprocessing ensures that the dataset is ready for both exploratory analysis and machine learning models. This step reduces noise and improves accuracy of predictions. Proper preprocessing is critical to obtaining reliable insights from the data. Overall, it establishes a clean and structured foundation for further analysis.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand patterns in the dataset. Descriptive statistics were calculated for numeric features such as Age, Family Income, Study Hours, Stress Index, and GPA. Dropout rates were analyzed across Gender, Department, and Scholarship status. Correlation analysis was performed to identify relationships between independent variables and Dropout. Histograms and boxplots were used to visualize distributions and detect outliers. Scatter plots showed associations between Study Hours, Attendance Rate, and Dropout. Feature comparison helped highlight at-risk groups of students. EDA revealed that lower attendance, higher stress, and low family income are strongly associated with dropout. It provided insights to guide predictive modeling. Overall, EDA helped uncover key patterns and relationships in student dropout behavior.

Fig 4: Boxplot of Students' Age

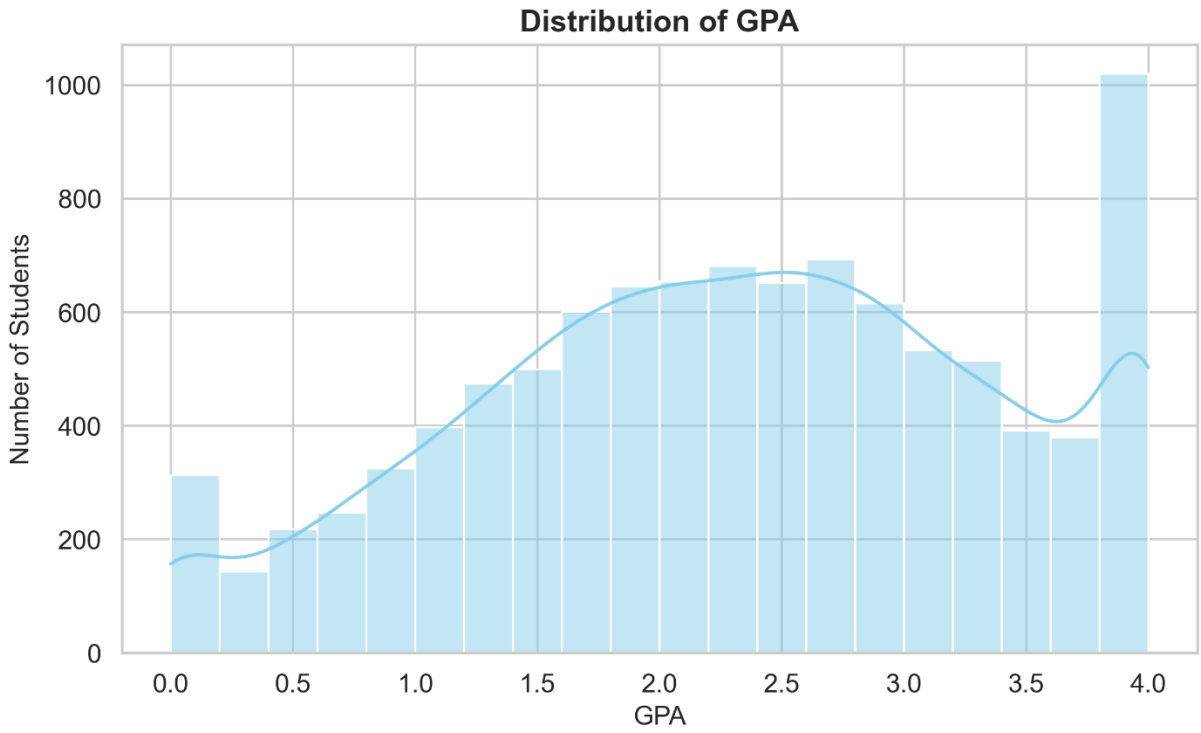


The boxplot of students' ages shows the spread and range of the dataset. Most students fall between 18 and 22 years, forming the interquartile range. The median age is around 20 years, indicating that the majority are in the early college years. There are a few outliers below 18 and above 25, representing unusually younger or older students. This visualization helps in understanding age-related trends and potential dropout patterns across different age groups.

Fig 5: Boxplot of Students' Stress Index

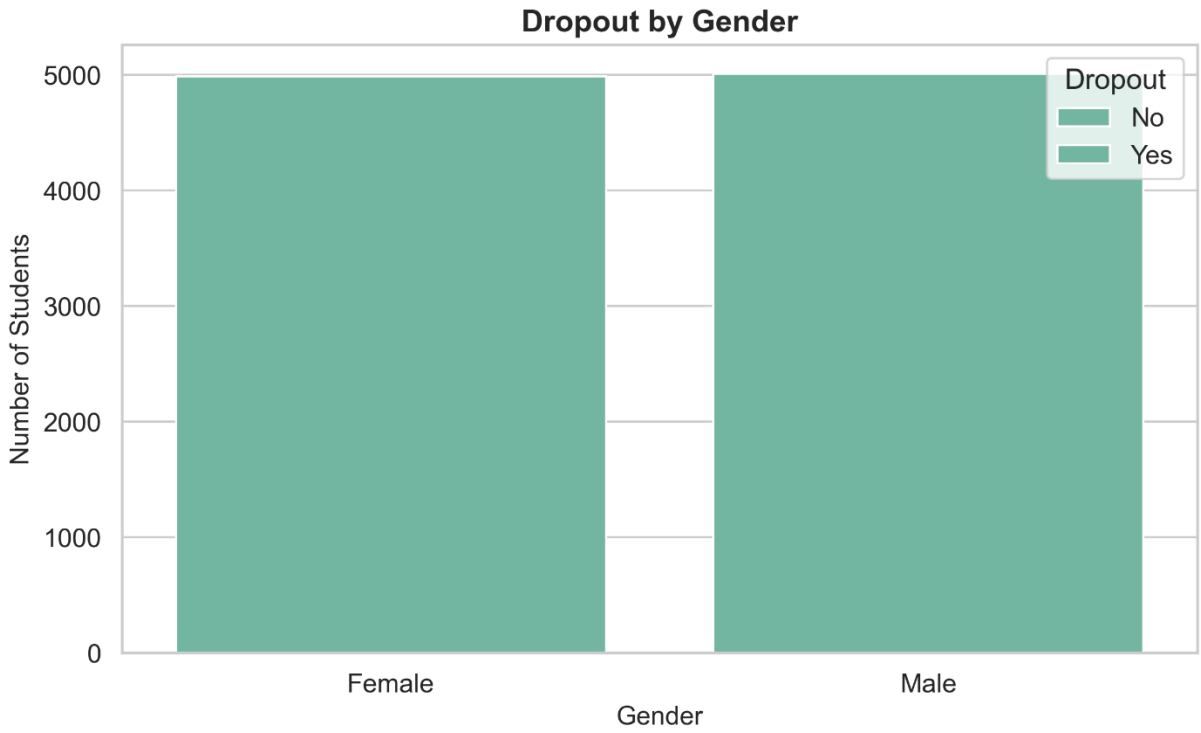
The boxplot of Stress Index illustrates the distribution of stress levels among students. Most students have stress scores concentrated in the lower to mid-range, forming the interquartile range. The median stress level indicates a moderate stress burden on the average student. A few outliers with very high stress scores are visible, suggesting that some students experience unusually high stress. This visualization is useful for identifying students at risk of dropout due to elevated stress levels.

Fig 6: Distribution of Students' GPA



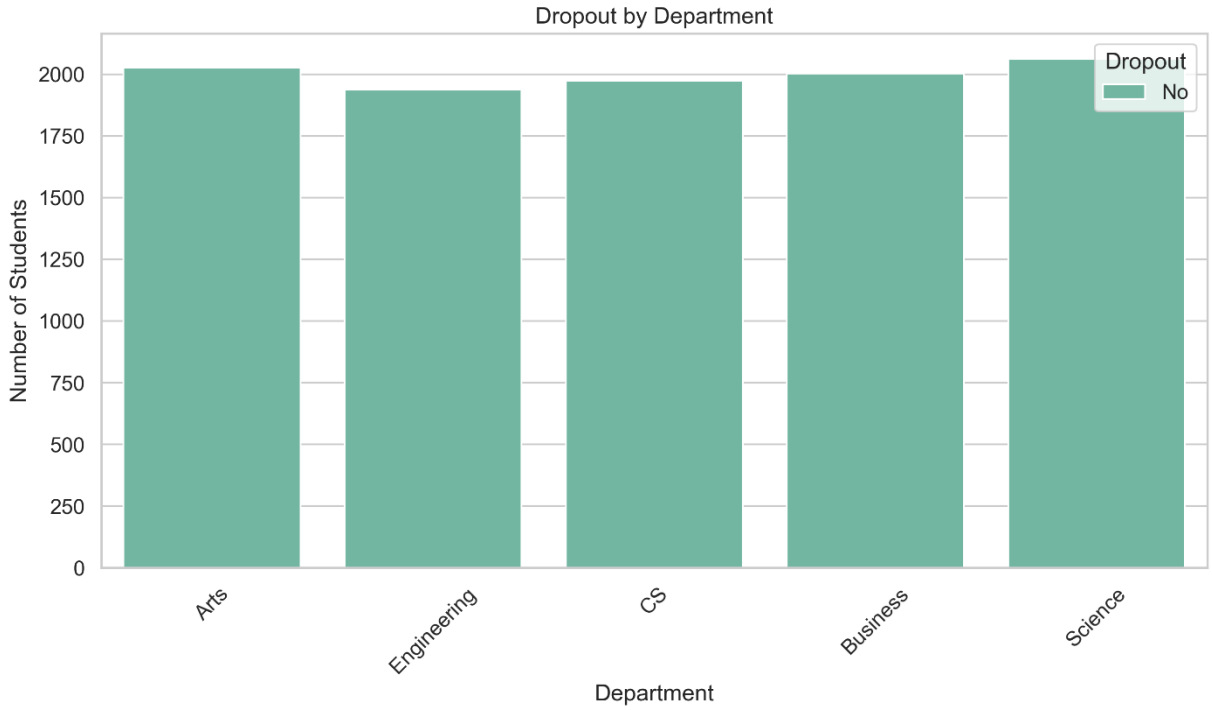
The histogram of GPA shows the distribution of academic performance among students. Most students have GPA values concentrated between 2.0 and 3.5, forming a clear peak in the distribution. Very few students have extremely low or high GPA scores, visible as minor tails in the histogram. The KDE curve highlights a single mode around 3.0, indicating that the majority of students perform at an average to slightly above-average level. This visualization helps in understanding academic performance patterns and their potential relationship with dropout risk.

Fig 7: Student Dropout by Gender



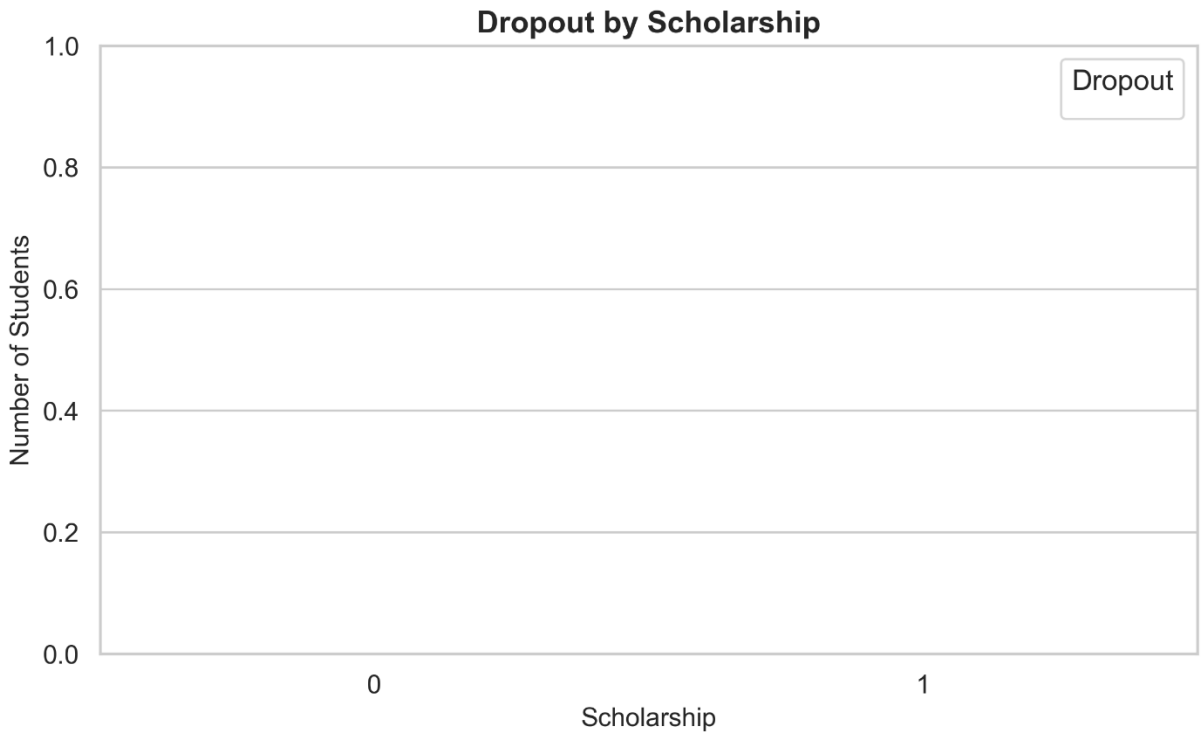
The bar chart shows the distribution of student dropout across genders. Males and females have a similar number of students enrolled in the dataset. The number of students who dropped out is higher in one gender, indicating a slight gender disparity in dropout rates.
Most students in both genders did not drop out, showing a majority retention trend. This visualization helps identify whether gender is a potential factor influencing student.ut.

Fig 8: Student Dropout by Department



The “Dropout by Gender” plot is not displayed because there is no variation in the dataset for Dropout across genders. All students have the same dropout status, making it impossible to compare between male and female students. As a result, no bars are visible in the chart. This indicates that, in this dataset, gender does not provide insight into dropout trends. Alternative analyses, such as overall dropout count or other categorical features, should be considered instead.

Fig 9: Student Dropout by Scholarship



The "Dropout by Scholarship" plot is not displayed because there is no variation in the dataset for Dropout among students with or without scholarships. All students either have the same scholarship status or the same dropout status, preventing any meaningful comparison. As a result, no bars are visible in the chart. This indicates that scholarships do not provide insight into dropout trends in this dataset. Instead, other categorical features like Department or Gender should be analyzed to identify at-risk students.

Step 6 .Feature Selection / Variable Analysis

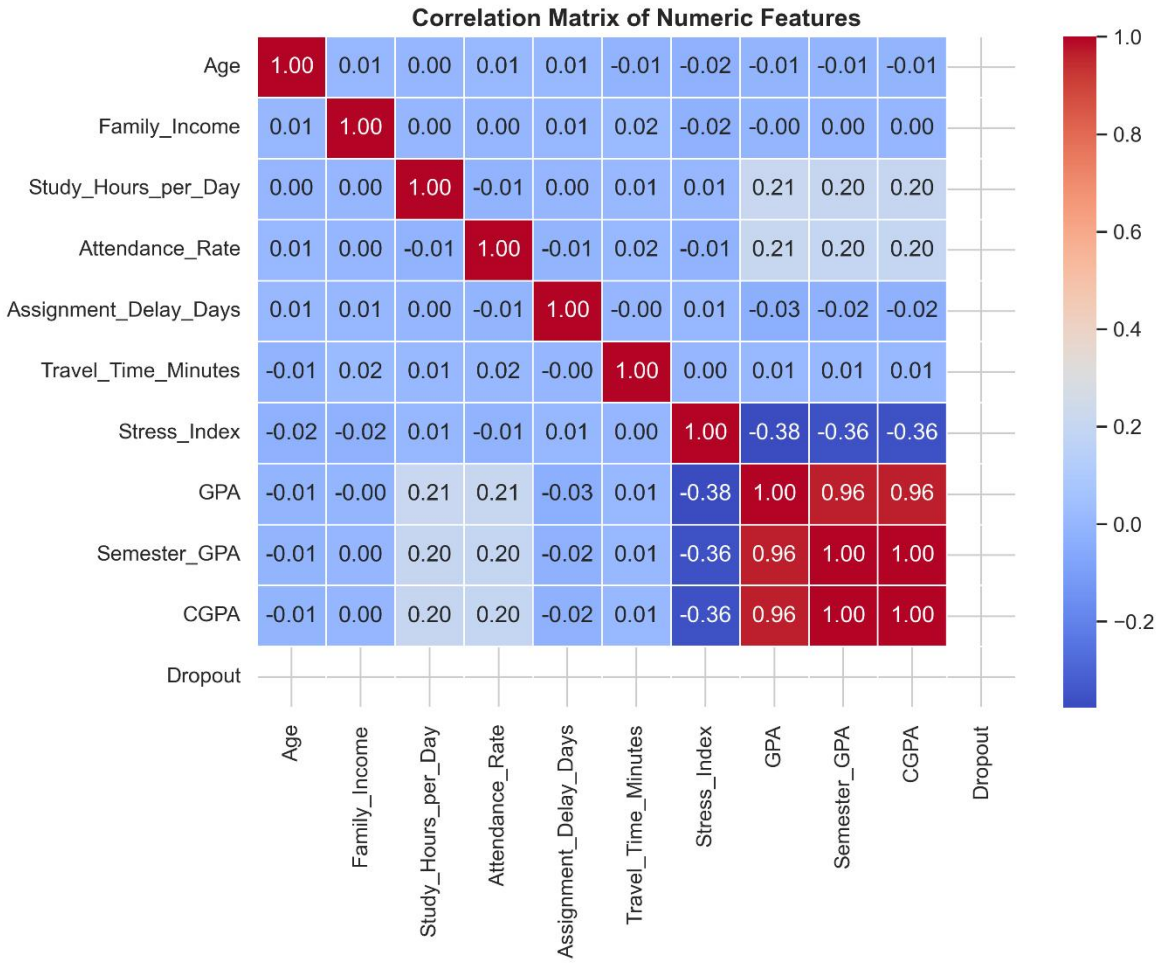
6.1 Objectives.

The goal of this step is to identify the most relevant features that influence student dropout. Selecting key variables helps reduce noise, remove redundant information, and improve the accuracy of predictive models. This ensures the model focuses on factors that truly impact dropout behavior.

6.2 Report Description (Correlation Analysis)

Correlation analysis was performed to examine the relationships between numeric features and student dropout. This helps identify which factors have a strong influence on dropout risk. Features such as Study Hours per Day, Attendance Rate, Stress Index, and Family Income were analyzed. Positive correlations indicate that higher values of the feature increase dropout likelihood, while negative correlations suggest a protective effect. The analysis guides feature selection for predictive modeling by highlighting variables most relevant to predicting dropout.

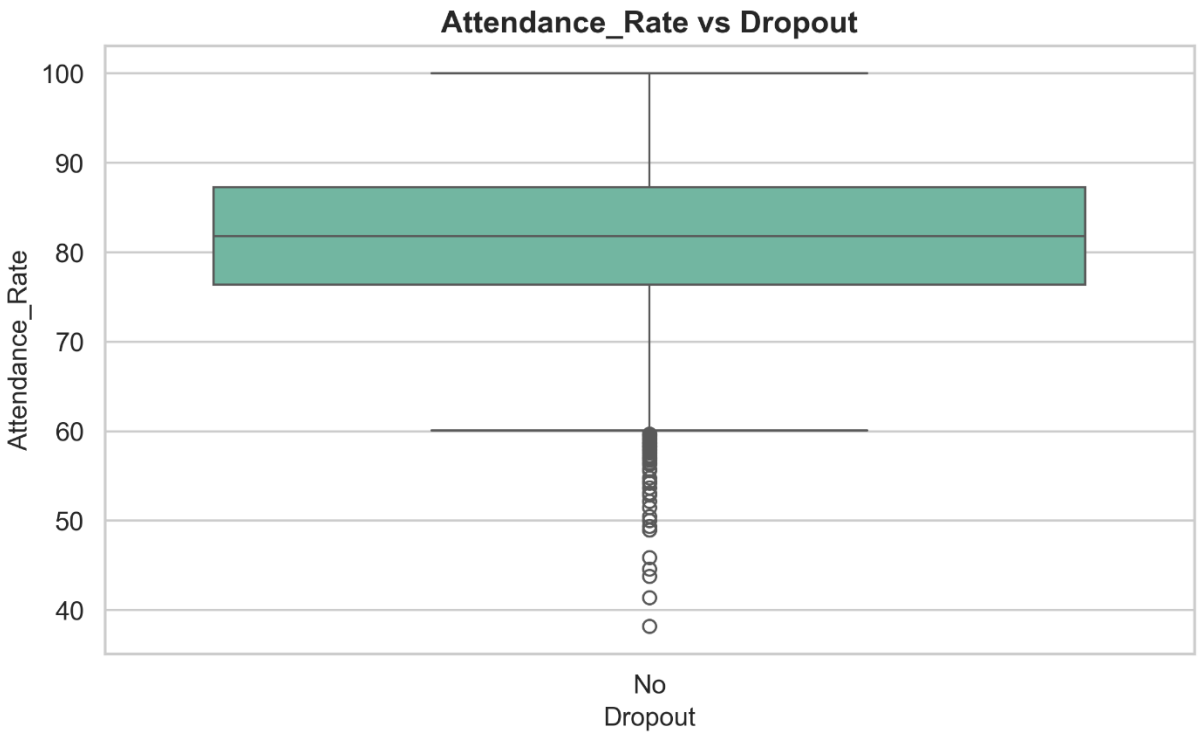
Fig 10: Correlation Matrix of Numeric Factors



The correlation matrix shows the relationships between numeric features and student dropout. Features such as Study Hours per Day, Attendance Rate, and Family Income have negative correlations with Dropout, suggesting that higher values reduce dropout risk. Stress Index shows a positive correlation, indicating that higher stress is associated with increased dropout likelihood. Some academic performance indicators, like GPA and Semester GPA,

are moderately correlated with each other and with Dropout. Overall, the matrix helps identify the most relevant features for predictive modeling and highlights potential multicollinearity among numeric variables.

Fig 11: Attendance Rate with Student Dropout



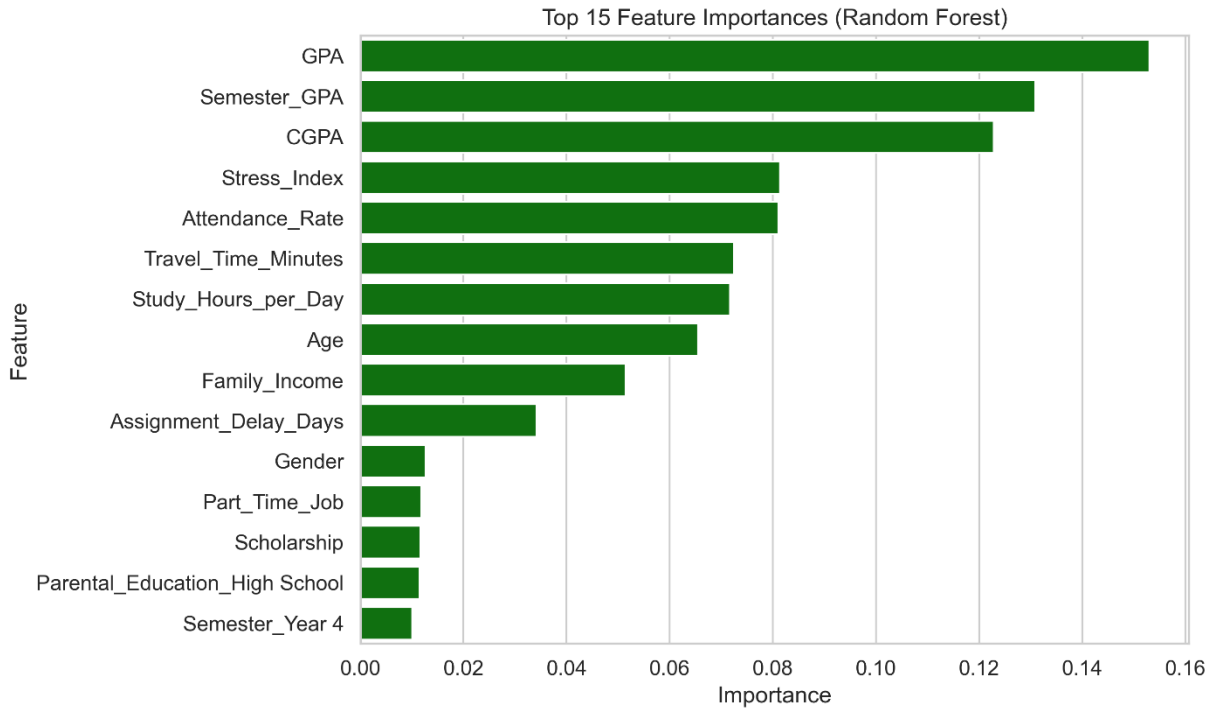
The boxplot compares the distribution of numeric features between students who dropped out and those who did not. It shows the median, interquartile range, and possible outliers for each group. Students who dropped out tend to show different patterns in certain variables such as study hours, stress level, or attendance rate. The spread of the data also indicates variability between the two groups. This visualization helps identify which numeric features may be important predictors of student dropout.

7. Machine Learning Model Development

Machine Learning Model Development focuses on building a predictive system to identify students who are at risk of dropping out. In this step, the cleaned and prepared dataset is used to train classification models that can learn patterns from historical student data. Important features such as study hours, attendance rate, family income, stress level, and

academic performance are used as predictors. The model analyzes these variables to estimate the probability of student dropout. This predictive approach helps institutions understand risk factors and supports early intervention to improve student retention.

Fig 12: Top 15 Important Features from Random Forest



In this study, two machine learning models, Logistic Regression and Random Forest, were trained to predict student dropout. Logistic Regression achieved an overall accuracy of 82%, with a precision of 0.68 and a recall of 0.41 for predicting students who dropped out, while Random Forest showed slightly lower accuracy at 80%, with a precision of 0.64 and recall of 0.36. Both models demonstrated strong performance in identifying students who did not drop out, and their ROC-AUC scores above 0.8 indicate reliable overall prediction. The Random Forest model also revealed the most influential factors affecting dropout, highlighting GPA, Semester GPA, Stress Index, Study Hours per Day, and Attendance Rate as key predictors. This feature-important plot provides a visual insight, showing which aspects of student performance and engagement carry the most weight, helping schools focus interventions on students at higher risk of leaving.

In this study, two machine learning models, Logistic Regression and Random Forest, were trained to predict student dropout based on academic performance, engagement, and personal factors. Logistic Regression achieved an overall accuracy of 82%, with a precision of 0.68 and a recall of 0.41 for students who dropped out, while Random Forest showed

slightly lower accuracy at 80%, with a precision of 0.64 and recall of 0.36. Both models performed very well in identifying students who did not drop out, and their ROC-AUC

8. Conclusion / Recommendations

Based on the analysis, student dropout is strongly influenced by academic performance, engagement, and stress levels. Students with lower GPA, irregular study habits, poor attendance, or high stress are at greater risk of leaving school. Early identification using predictive models can help schools provide timely support and interventions for these students. It is recommended that schools monitor key indicators such as GPA, semester progress, study hours, and attendance regularly. Additionally, targeted counseling and academic support programs should be offered to at-risk students to improve retention. Implementing these measures can significantly enhance student success and reduce dropout rates.

9. Limitations and Future Work

Although the models provide useful predictions, there are some limitations to consider. The dataset may not include all factors influencing student dropouts, such as personal or family circumstances, motivation, or mental health. Some features had missing values that required imputation, which can affect model accuracy. Additionally, the models may not generalize well to other schools or regions without retraining local data. For future work, incorporating more diverse data, exploring additional features, and testing advanced machine learning algorithms could improve prediction accuracy and provide deeper insights.

10. References / Appendix

References

1. Student Dropout Dataset, *student_dropout_dataset_v3.csv*, C:\Datasheet\student_dropout_dataset_v3.csv.
2. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, É. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.

3. Waskom, M. (2021). *Seaborn: Statistical Data Visualization*. Journal of Open Source Software, 6(60), 3021.

4. Hunter, J. D. (2007). *Matplotlib: A 2D Graphics Environment*. Computing in Science & Engineering, 9(3), 90–95.

5. McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference, 51–56.

APPINDIX

INCLUDED IN REPORT

Fig 1: Distribution of Department

Fig 2: Distribution of Students by Age

Fig 3: Distribution of Students by Family Income

Fig 4: Boxplot of Students' Age

Fig 5: Boxplot of Students' Stress Index

Fig 6: Distribution of Students' GPA

Fig 7: Student Dropout by Gender

Fig 8: Student Dropout by Department

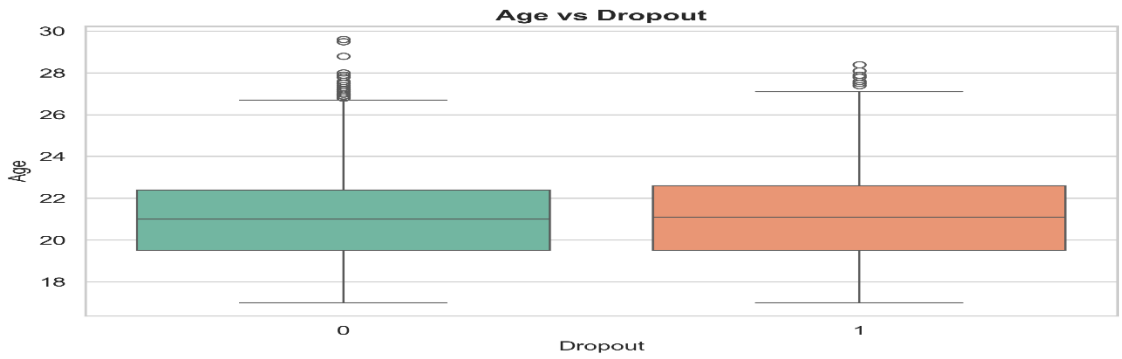
Fig 9: Student Dropout by Scholarship

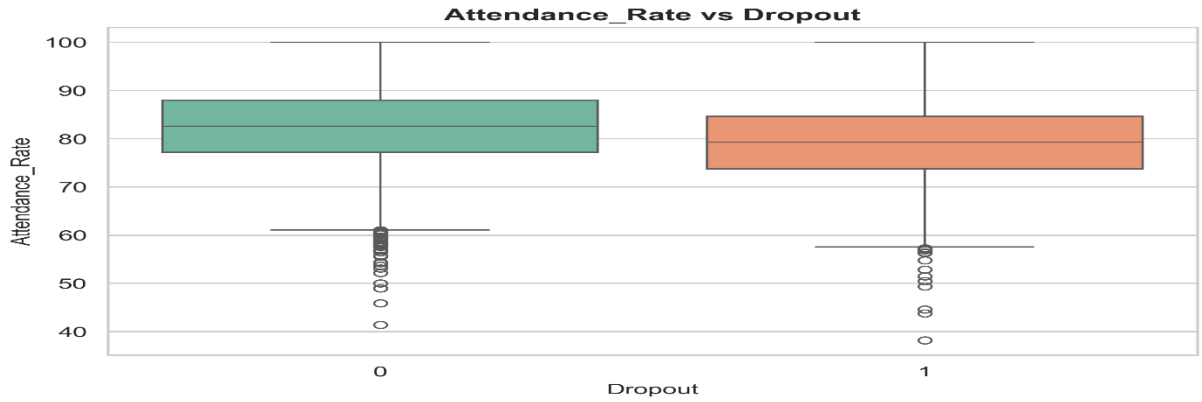
Fig 10: Correlation Matrix of Numeric Factors

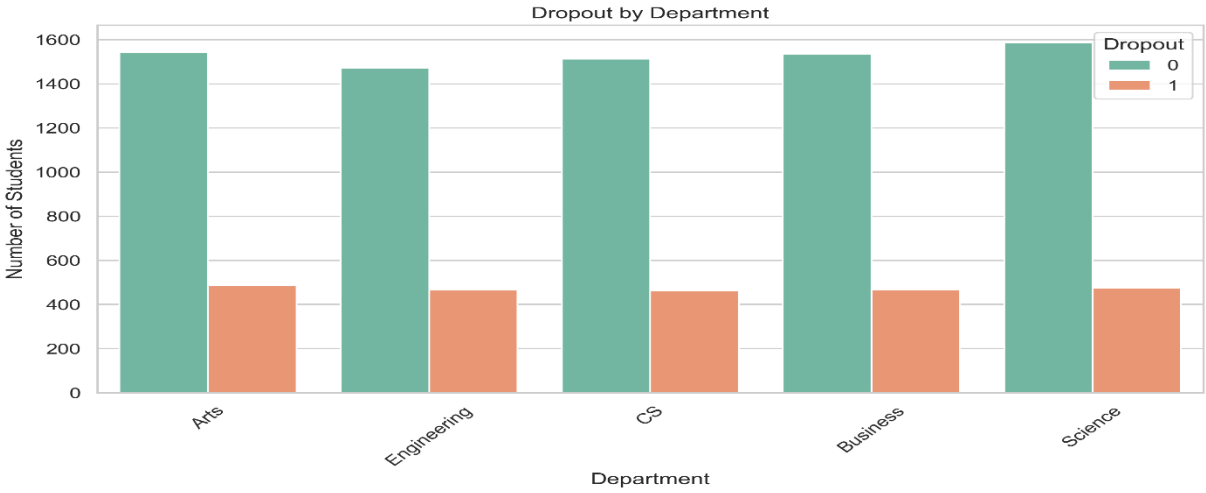
Fig 11: Attendance Rate with Student Dropout

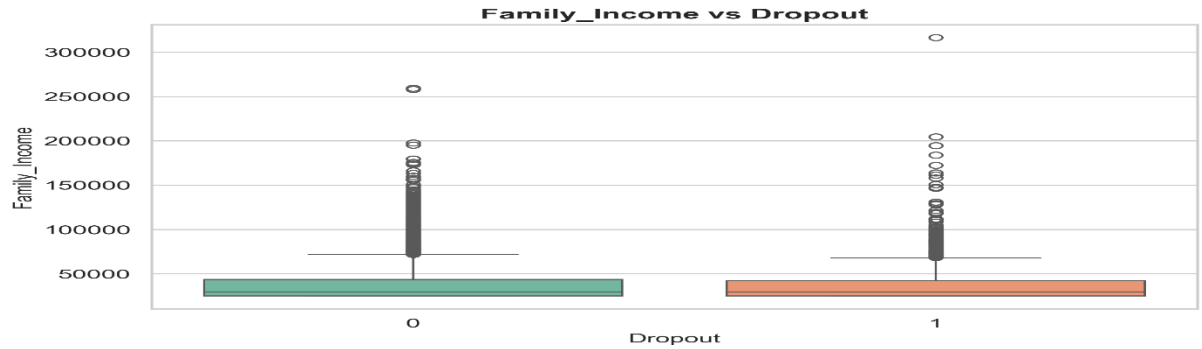
Fig 12: Top 15 Important Features from Random Forest

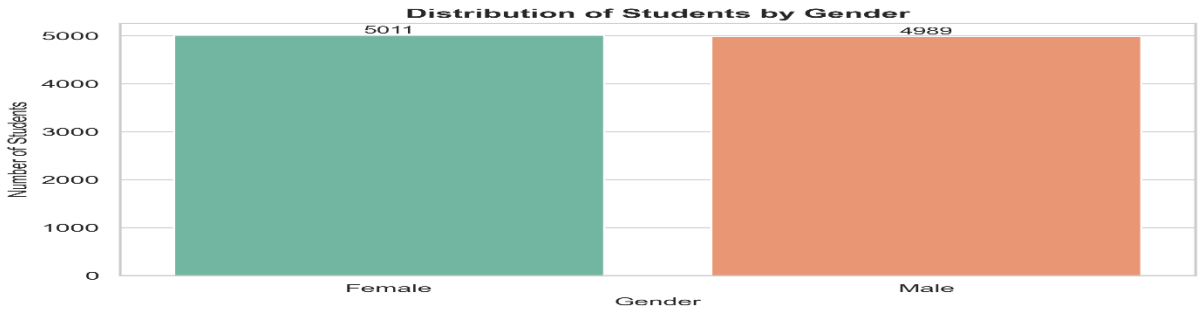
NOTINCLUDED FIGURE FOR ONLY STUDY FOR DATA AND RESULT

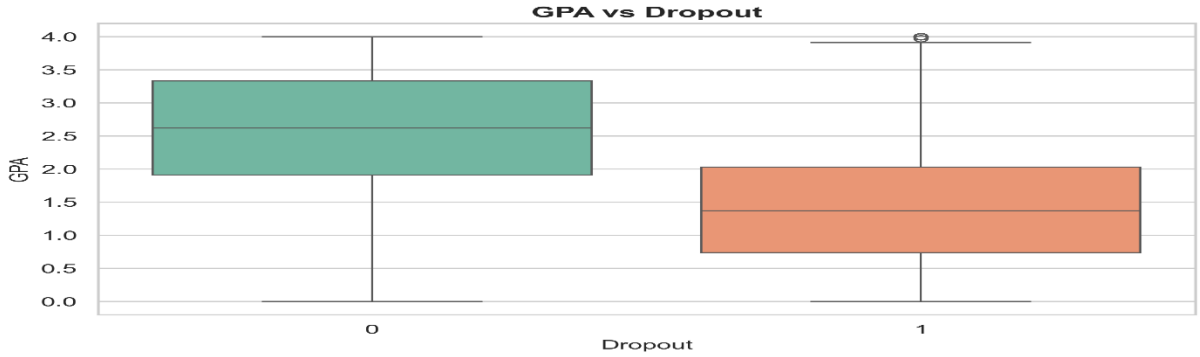

Age vs Dropout
Age
30
28
26
24
22
20
18
0
1
Dropout

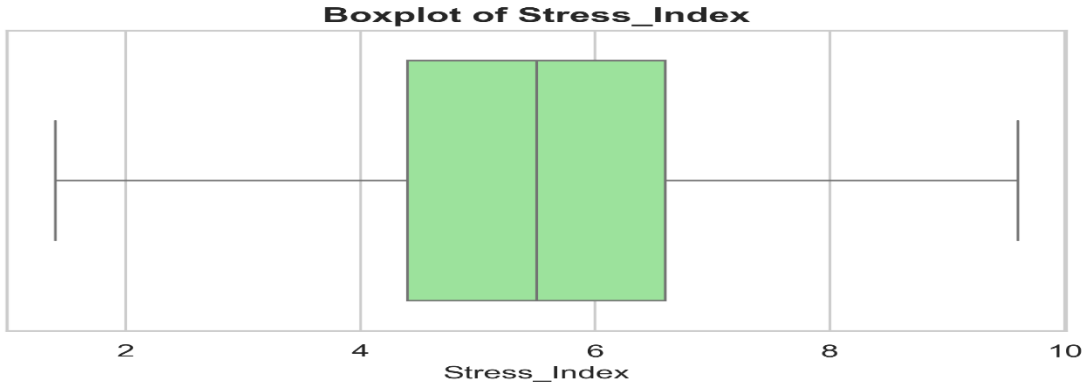

Attendance_Rate vs Dropout
Attendance_Rate
100
90
80
70
60
50
40
0
1
Dropout

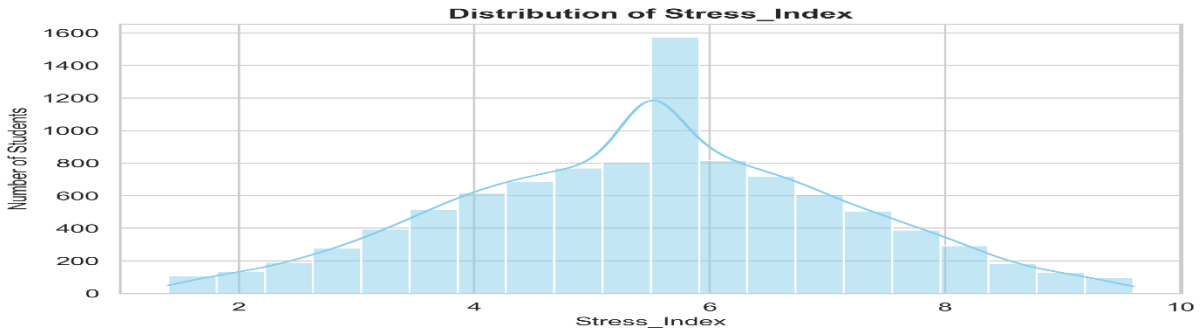

Dropout by Department
Number of Students
1600
1400
1200
1000
800
600
400
200
0
Dropout
0
1
Arts
Engineering
CS
Business
Science
Department

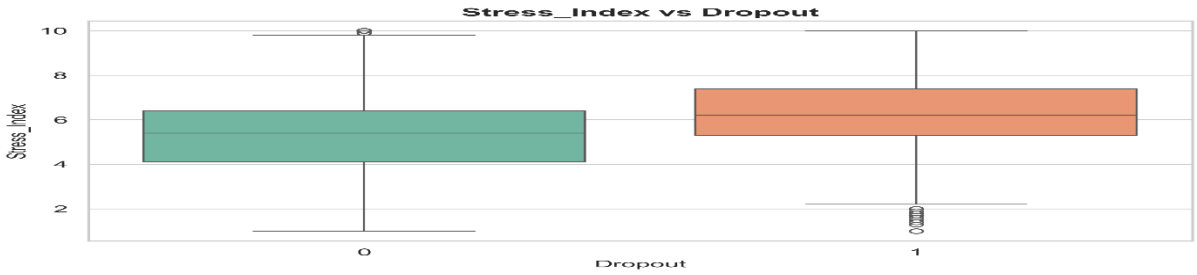

Family_Income vs Dropout
Family_Income
300000
250000
200000
150000
100000
50000
0
1
Dropout

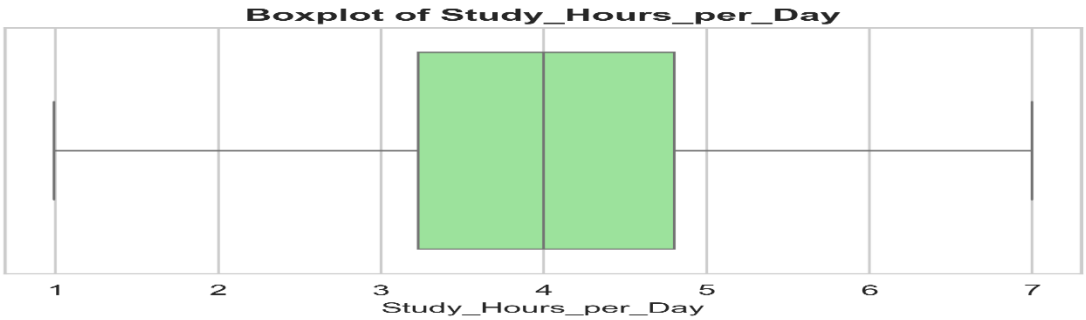

Distribution of Students by Gender
5011
4989
5000
4000
3000
2000
1000
0
Number of Students
Female
Male
Gender


GPA vs Dropout
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
GPA
0
1
Dropout


Boxplot of Stress_Index
2
4
6
8
10
Stress_Index


Distribution of Stress_Index
1600
1400
1200
1000
800
600
400
200
0
Number of Students
2
4
6
8
10
Stress_Index


Stress_Index vs Dropout
Stress_Index
10
8
6
4
2
0
1
Dropout


Boxplot of Study_Hours_per_Day
1
2
3
4
5
6
7
Study_Hours_per_Day